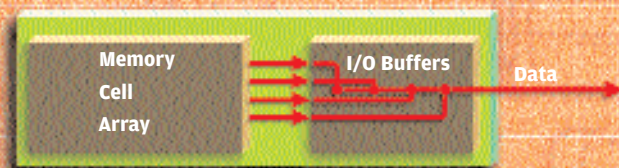


DDR 2 Memory Multiplexing



DDR 2 data flow diagram: Four tiers of multiplexing at work, I/O Buffer speed is also twice the DRAM frequency, unlike DDR RAM, which is synchronous

understand. First we look at the core frequency/clock, which represents the actual speed of the DRAM chips. In the case of DDR 200 MHz memory, the core speed is 100 MHz; in the case of DDR 400 memory, it is 200 MHz.

Then we have the clock frequency, also called frequency of the I/O buffers. For DDR memory, the I/O buffer speed is synchronous with the DRAM frequency. However, for DDR2 memory, the I/O buffer speed is twice that of the core. Therefore the I/O buffers need to be supplied with twice the amount of data that the DDR memory buffers require, to avoid serious latency issues. This is exactly why the multiplexing level in the case of DDR 2 memory has increased to four, twice that of DDR. In the case of DDR memory at a core speed of 200 MHz, your I/O buffer speed is also 200 MHz, and the data speed becomes 400 MHz. In the case of DDR2 400 MHz memory, the data frequency and the I/O buffer remain identical, at 400 MHz and 200 MHz respectively. What changes is the core speed. It's down to 100 MHz! Remember what we said about the relationship between core speeds and I/O buffer speeds? Well, for a core speed of 200 MHz, the I/O buffer frequency becomes

400 MHz, and the data frequency becomes 800 MHz.

Looking at these figures, the difference is glaring—at the same core/DRAM speeds, the bandwidth effectively doubles!

The Role Of Latency

One of the most important characteristics of memory and a significant performance-affecting factor, is latency. Latency can be quite simply defined as wasted clock cycles. It occurs since DRAM memory cells have to continually refresh themselves. Latency, therefore, basically represents delay; it's the time taken for the memory to get ready for a fetch or deliver data transaction. There is also a certain amount of unavoidable latency that occurs between the activation of a column or row, due to the time required to set up the addresses of the same. Note that latency is an omnipresent factor; it can only be minimised, and never completely done away with.

Now latency is directly proportional to clock speeds, so when a MHz bump occurs, the latency figures also rise. Let's compare a DDR 400 and a DDR2 400 MHz module. Typical latencies would be 2.5-3-3-6 for a DDR 400 MHz module, while a DDR2 400 MHz would only manage 3-4-4-8, making it slower as far as data accessibility goes. So DDR 2 scores bandwidth-wise, but loses out latency-wise. A paradoxical situation, with only one solution—bump up the clock speeds further. And DDR2 has done exactly that, delivering 667 MHz and 800 MHz, with 1066 MHz promised in the near future. With AMD 64's efficient memory controller and Intel's Conroe's bumped up FSB speeds (1066 and 1333 MHz), this extra bandwidth will go down real easy!

What's New... About DDR2?

In addition to the additional bandwidth DDR2 provides, and its superior technology (TSOP is inferior to the FBGA packaging), savings in

RAM Latencies—What All Those Figures Mean

So we know that latency-wise, DDR2 is slower than DDR memory. Latency figures are measured as 3-3-3-8, or something similar, and most users are lost when it comes to the meaning of such figures.

The figure 3-3-3-8 represents delay. The smaller these values can get, the faster the memory is (clock speeds remaining constant, of course). It's worth noting that these figures represent the minimum latency, meaning the timings cannot get any smaller, that is, tighter. DDR 400 MHz memory at timings of 2-3-2-6 would, for example, be much faster than similar memory at 3-3-3-8 timings, simply because the latency is reduced. Reduced latency leads to what is called "tighter timings" by enthusiasts.

The first figure in the above example is the most significant speed-wise. It represents the CAS or Column Address Strobe latency figure.

As we know, RAM first has to read a command sent to it, and then output some data based on that command. The CAS represents the delay between a registered read and data output. It's measured in clock cycles. Therefore a CAS latency value of 3 means three clock cycles will complete

before data is ready to be sent forward. CAS Latency is often abbreviated as CL.

The second figure (the second 3 in the example) represents the RCD or Row - Column Delay. It is defined as the number of clock cycles required between RAS and CAS. As latency, it's the delay between defining a row and column in a particular memory block, and the read/write operation to that particular location.

RP is the Row Pre-charge time. It's denoted by the third 3 in our example. In memory, each row in the bank needs to be closed, that is, terminated before the next row can be accessed. The RP represents, in clock cycles, the time needed to terminate and open a row of memory (open being the current state), and to access the next row.

RAS stands for Row Address Strobe. This is the last number in our example, i.e. 8. There is a delay between requesting of data and the actual issue of a pre-charge command. This difference is basically the amount of clock cycles spent in order to access a certain row of data in memory. This delay between data request and pre-charge is called RAS or active to pre-charge delay.